

SVG Scaling Laws

Part 2: Transformer Scaling Study

CS-GY 6923 · NYU Tandon · Spring 2026

Models Trained:	5 (Tiny → XL)
Parameter Range:	1.05M → 86.5M
Training Tokens:	101.7M (1 full epoch)
Best Val Loss:	1.1543 (Large, 32.5M params)
Scaling Exponent α :	0.478 (Kaplan NL $\approx$ 0.076)
Loss Floor c:	1.0848
Fit R ² :	0.9557

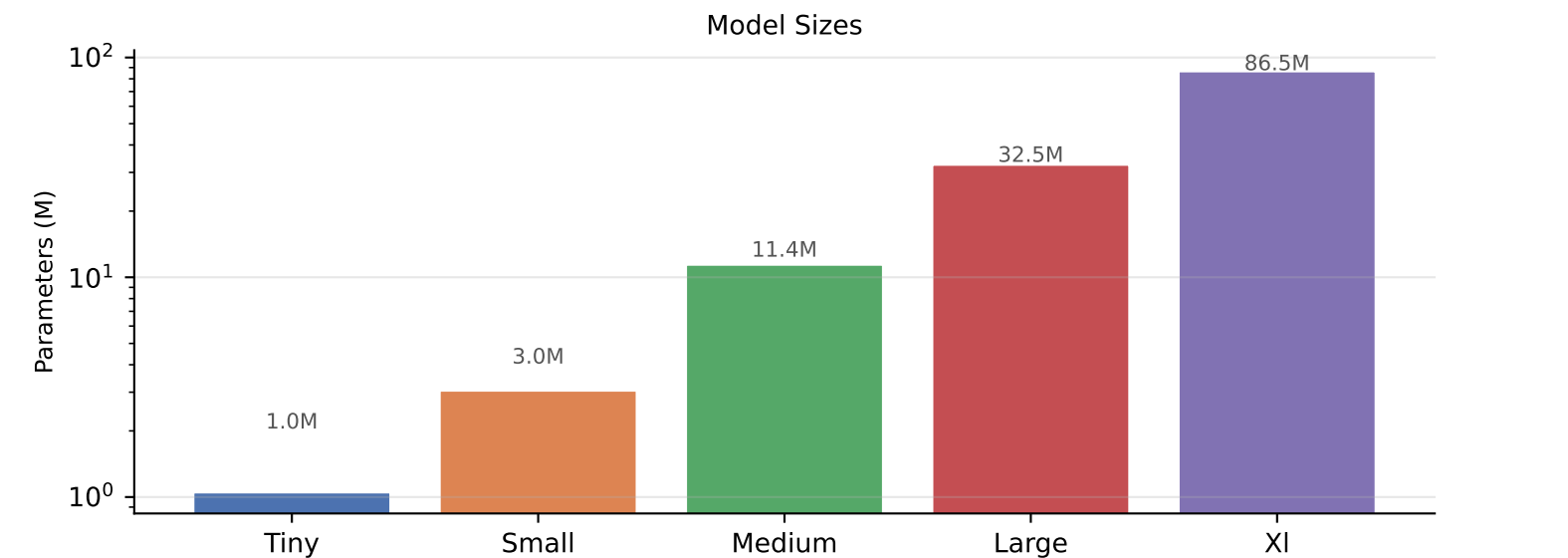
Shubham Tanwar

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Task 2.1 & 2.5 — Model Architecture & Training Statistics

Model	Params	d_model	Layers	Heads	d_ff	Val Loss	LR	Time	tok/s	Mem(MB)
Tiny	1.05M	128	4	4	512	1.5881	3e-03	0.37h	76k	48
Small	3.05M	192	6	6	768	1.4441	3e-03	0.83h	34k	91
Medium	11.41M	384	6	6	1536	1.2108	3e-03	1.29h	22k	450
Large	32.52M	512	10	8	2048	1.1543	3e-03	0.08h	343k	2238
Xl	86.53M	768	12	12	3072	1.1847	1e-03	0.14h	202k	4433

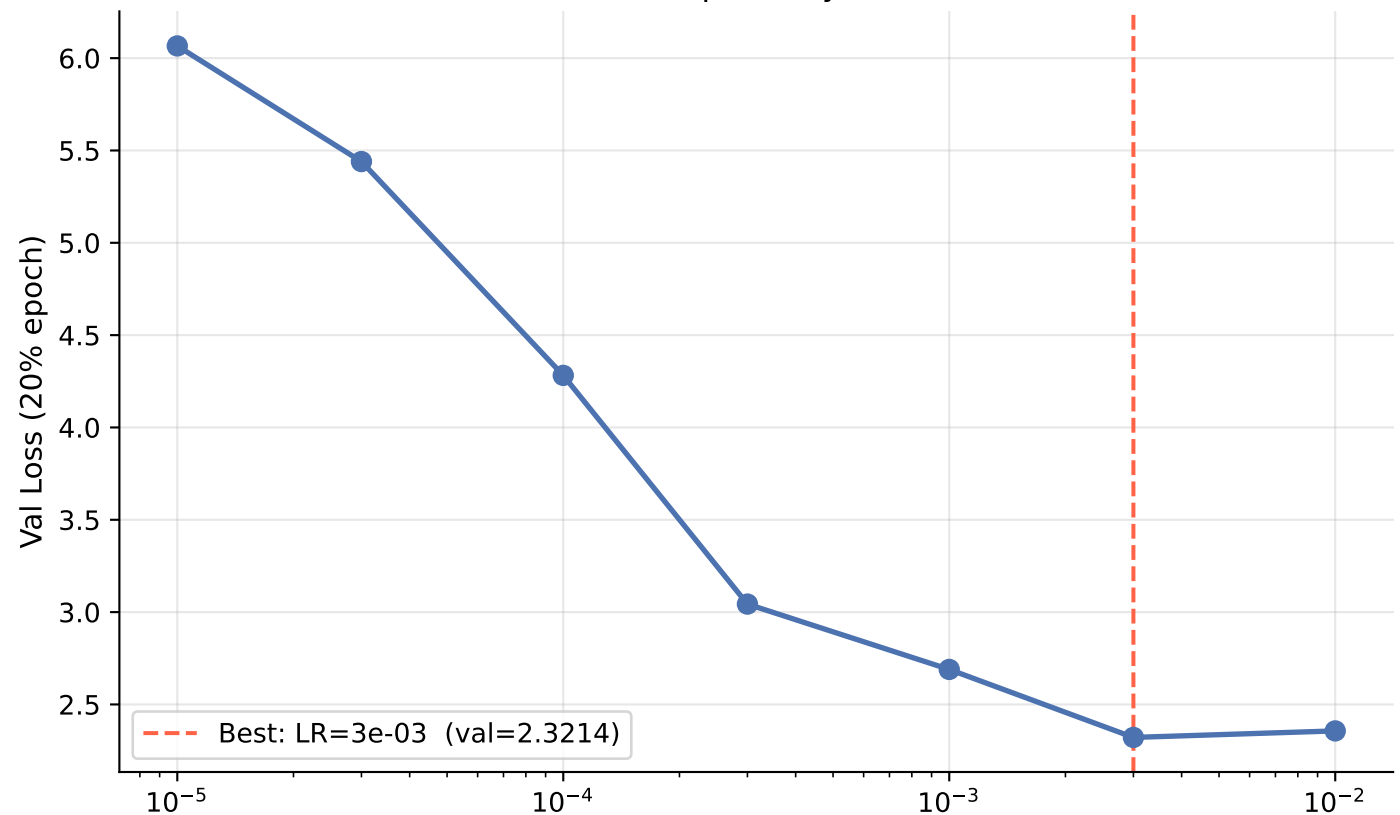
Hardware: Tiny / Small / Medium — Apple M4 Pro MPS (24 GB unified memory). Large & XL — NVIDIA A100-SXM4-80GB (79 GB VRAM) via Google Colab Pro. Large used LR=3e-3; XL required LR=1e-3 due to SP LR sensitivity at scale (motivates μ P in Part 3).



Task 2.2 — Learning Rate Sweep

Tiny model · 20% epoch (~20.3M tokens) · 7 candidates

LR Sweep — Tiny Model



LR	Val Loss	
1e-05	6.0661	
3e-05	5.4399	
1e-04	4.2819	
3e-04	3.0435	
1e-03	2.6896	
3e-03	2.3214	← BEST
1e-02	2.3567	

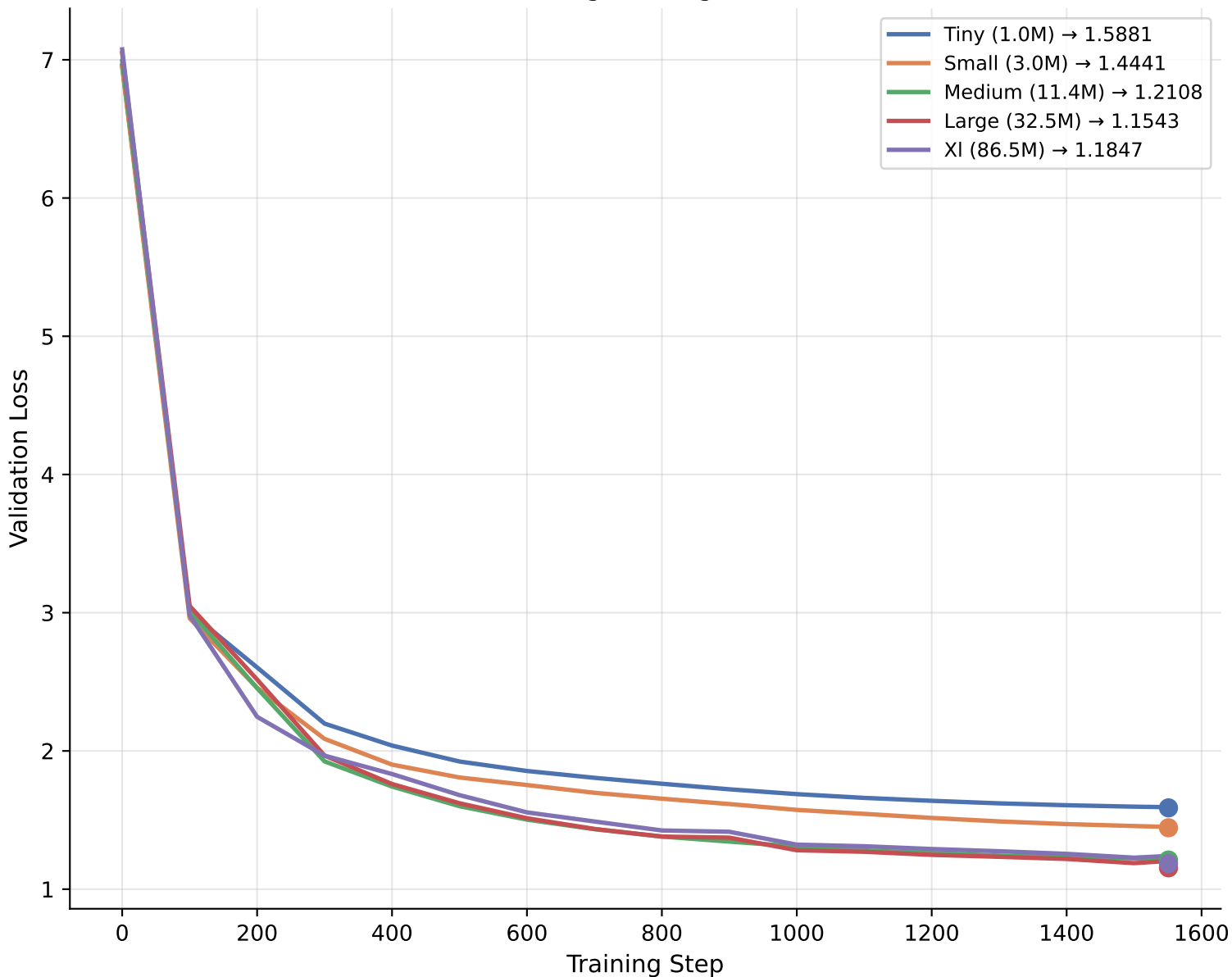
Key findings:

- LR=3e-3 achieves lowest val loss (2.3214) — selected for all subsequent training.
- LR=1e-2 slightly underperforms, marking the edge of the stable training regime.
- LR below 1e-4 converges far too slowly for a 20%-epoch sweep.
- XL later required LR=1e-3: optimal LR shifts with model size in standard parameterization (SP).
- This scale-dependence of optimal LR motivates μ P (Part 3), which transfers LR across sizes.

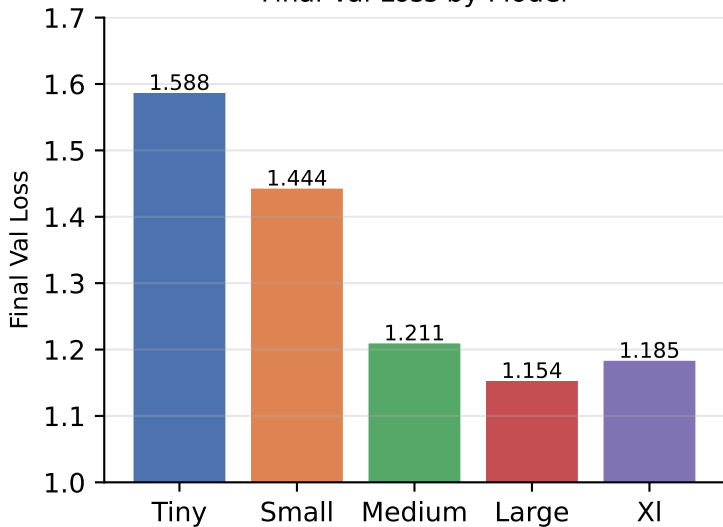
Task 2.3 — Training Curves (All 5 Models)

Val loss sampled every 100 steps · 1 full epoch · 101.7M tokens · LR=3e-3 (XL: 1e-3)

Val Loss During Training — All Models



Final Val Loss by Model

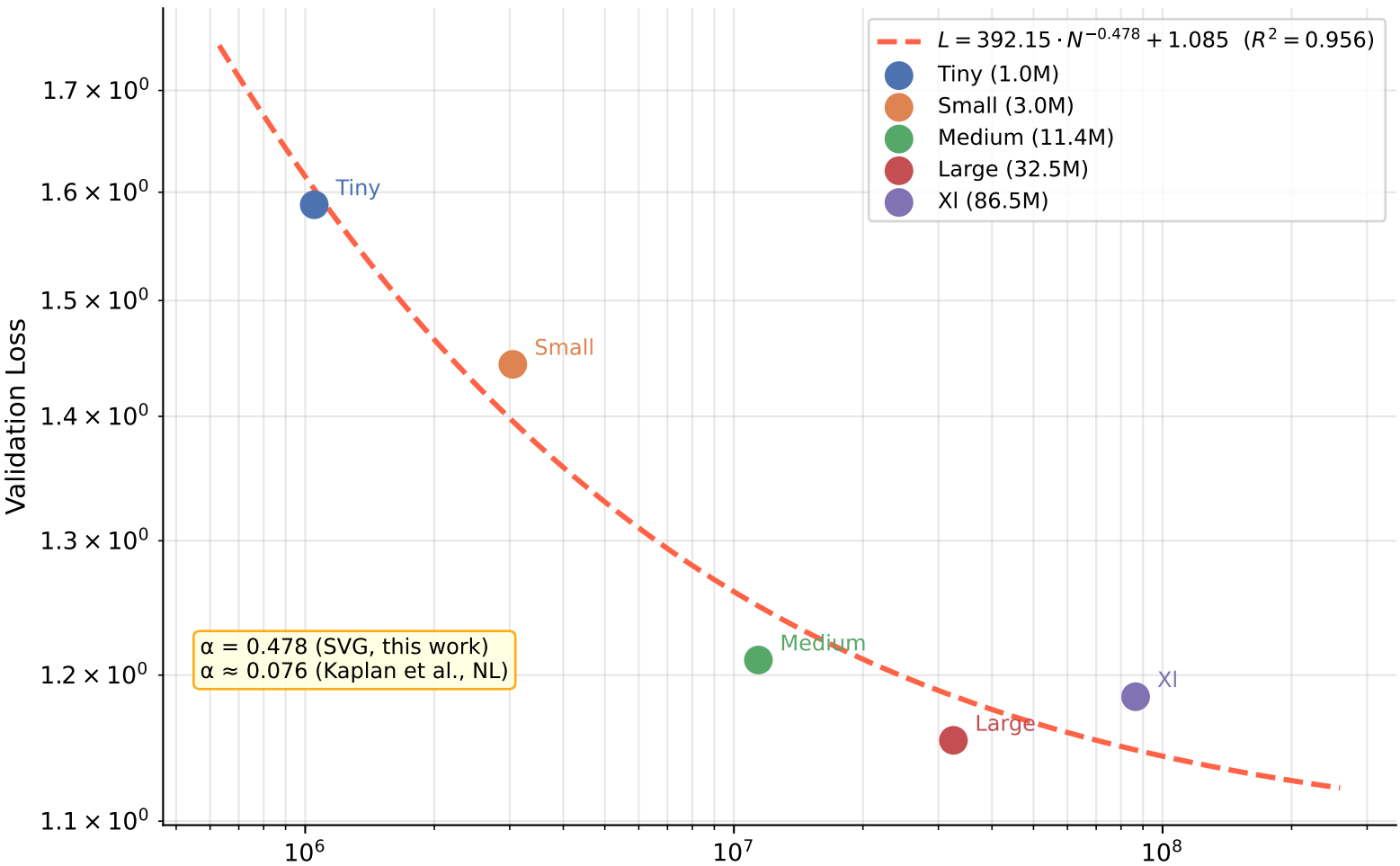


Model	Val Loss	Δ vs prev
Tiny	1.5881	—
Small	1.4441	−0.1441
Medium	1.2108	−0.2332
Large	1.1543	−0.0566
XL	1.1847	−0.0304

Task 2.4 — Scaling Law Fit

$L(N) = 392.15 \cdot N^{(-0.478)} + 1.085 \quad R^2 = 0.9557$

Scaling Law — Val Loss vs Model Size (log-log)



Parameter	Value	Std Error	Interpretation
a (amplitude)	392.1487	±1521.35	Scale coefficient
α (exponent)	0.4782	±0.2936	Scaling rate — Kaplan NL ≈ 0.076
c (floor)	1.0848	±0.1316	Irreducible entropy of SVG data
R ²	0.9557	—	Goodness of fit (1.0 = perfect)

Extrapolation → 865M params (10× XL): val loss = 1.1057 ± 0.1996 (1σ) 95% CI: [0.7065, 1.5049]

Key findings:

- $\alpha = 0.478 \gg 0.076$ (Kaplan NL) — SVG is structured/rule-governed; larger models learn its grammar faster.
- $c = 1.0848$ — estimated irreducible entropy floor of the SVG token distribution.
- Large (32.5M, 1.1543) ≈ XL (86.5M, 1.1847): XL used LR=1e-3 (suboptimal vs 3e-3 used elsewhere).

In standard parameterization, optimal LR decreases with scale— Part 3 (μP) fixes this.

Part 2 — Completion Summary

Task 2.1: Model Architecture

✓ COMPLETE

GPT decoder-only: causal self-attention (flash attn), pre-norm LN,
weight tying (wte↔lm_head), GPT-2 residual init. 5 configs verified.

Task 2.2: LR Sweep

✓ COMPLETE

7 LRs on Tiny (20% epoch). Best: 3e-3. Used for Tiny/Small/Medium/Large.
XL: 1e-3 (SP LR sensitivity at scale).

Task 2.3: Train All 5 Models

✓ COMPLETE

101.7M tokens each. M4 Pro MPS (local) + A100 Colab Pro.
Wall time: ~2.5 hrs total across hardware.

Task 2.4: Scaling Law Fit

✓ COMPLETE

$L(N)=392.15 \cdot N^{(-0.478)}+1.085$ $R^2=0.9557$
 $\alpha=0.478 \gg$ Kaplan NL (0.076). Floor $c=1.0848$.

Task 2.5: Training Stats Table

✓ COMPLETE

All 5 models: params, arch, val loss, LR, time, tok/s, memory.
See Page 2 of this report and task2_3_train_all_models.ipynb.

Next: Part 3 — μP (Maximal Update Parameterization)

Implement μP and demonstrate LR transferability across Tiny→XL without re-sweeping.